

Sagar Mandiya

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EDUCATION

Boston University

Boston, MA

Master of Science in Computer Science (STEM)

Jan 2024

Relevant Coursework: Artificial Intelligence, Machine Learning, Software Engineering, Analysis of Algorithms, Data Science with Python, Web Mining and Analytics

SKILLS

Python, C/C++, Java, SQL, TensorFlow, PyTorch, AWS, GCP, Git, Docker, FastAPI, HTML5, CSS, JavaScript, Linux

EXPERIENCE

Software Engineer Intern

Boston, MA

Ipser Lab

May 2024 – Present

- Developed React based components for a business website to reduce load times by 20% and improve user experience.
- Utilized Java to resolve login issues at the back-end to ensure consistent performance across 5 distinct platforms.
- Actively contributed in brainstorming ideas with the managing team to crucially improve the overall product performance.

Graduate Research Assistant

Boston, MA

Boston University - Department of Computer Science

Mar 2023 – Mar 2024

- Analyzed ER data using machine learning to identify patterns, reducing press-machine injuries by over 20%.
- Developed visualizations in Python and Tableau, reducing injury misclassification by 15% and improving prediction accuracy.
- Applied ML and dimensionality reduction to analyze 15,000+ ASD cases in a major longitudinal study on first language acquisition.

Computer Vision Engineer

TensorGo Technologies

Sep 2021 – Aug 2022

- Researched and evaluated OCR models (Keras OCR, Tesseract, OpenVino, EasyOCR) on images with varying backgrounds.
- Fine-tuned OCR models for better performance in tough lighting conditions, improving accuracy by 15%.
- Led team to build and deploy BERT-based language model pipeline from scratch, cutting recruiter hiring time by 25%.
- Built an end-to-end ASR, Voice Biometrics, and Diarization pipeline using the Nvidia NGC stack, achieving a 10% faster model.
- Collaboratively led projects, resolved issues, and strategically engaged with team members and stakeholders.

PROJECTS

AutoCode for Python | Python, PyTorch, NLTK, ONNX, GCP, Google BigQuery, SQL

- Trained transformer based GPT-2 based LLM model from scratch to auto-complete python code similar to GitHub CoPilot.
- Employed Google BigQuery to gather a large 160GB python code dataset from Github to be used to train the LLM.

Deepfake Detection | Python, TensorFlow, TensorRT, XceptionNet, MisoNet, ResNet, VGG-19

- Designed and implemented neural network models for detecting artificially generated fake images/videos with 73.5% average precision on a large dataset of size 470GB+.
- Conducted a comparative study of SOTA neural network models to determine performance for deepfake detection. [\[paper\]](#)

PUBLICATIONS

Analyzing Patterns of Injury in Occupational Hand Trauma Focusing on Press Machines: A Registry-Based Study and Machine Learning Analysis [\[ITISE\]](#)

May 2023 - May 2024

Critical period for first language acquisition may be shorter in autistic children than in typical children [\[paper\]](#)

Jul 2023 – Jan 2024

Deep Fake Detection : Survey of Facial Manipulation Detection Solutions [\[paper\]](#)

Jan 2021 - Jun 2021